

NEMANIA BOROVITS

Data Scientist

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Professional skills

Data Analysis:



Programming language: Python.

Data engineering: Apache Spark, Apache Kafka, Elastic Stack.

Data Visualization: Plotly Dash, Matplotlib, Seaborn, Kibana, Tableau, Power BI.

Machine learning: Scikit-learn, NumPy, Pandas, Snorkel.

Deep learning: TensorFlow, Keras, PyTorch.

NLP: NLTK, Hugging face Transformers, spaCy, gensim, t-SNE.

Web Crawling: BeautifulSoup.

Privacy Frameworks: Flower, Opacus, SEAL.

Certificates: CIPP/E (in progress).

Cloud Technologies:



Google Cloud

Heroku

Dataiku

Microsoft Azure

Azure Databricks

Back-end Development:



Data modeling: Microsoft SQL Server, MongoDB

ORM: nHibernate, Entity Framework.

Business Logic: asp .net MVC, asp .net (C#).

API: fastAPI, Flask, asp .net Web API (C#), .

Web Services: SOAP, RESTful (C#, Python).

Front-end development:



Web skills: HTML5, Javascript, jQuery, AJAX, JSON, CSS, Bootstrap.

JS Frameworks: React js, Knockoutjs, VueJs.

Summary

As a data scientist with a strong problem-solving orientation, I have accumulated five years of experience as a full-stack developer and data engineer in Greece before relocating to the Netherlands. My passion for data science led me to pursue a Master's in Data Science and Entrepreneurship, followed by a PhD focusing on Data Engineering and AI for Privacy in collaboration with the Jheronimus Academy of Data Science and KPN. This partnership aims to apply my privacy engineering research within KPN's operations.

Currently, in my final PhD year, I am compiling my research publications, writing my thesis, and exploring professional opportunities where I can apply my data science, machine/deep learning, AI and data governance skills to make a meaningful societal impact.

Professional Experience

15/10/2020 - Data Scientist

KPN

Data Analytics: At the KPN Data Science Lab, I apply Privacy by Design principles to data pipelines for both analytics and governance. Collaborating with internal teams, external companies, and the legal department, I ensure all projects meet GDPR compliance. My focus is on privacy-preserving NLP for analyzing Dutch call center text data. I work closely with subject matter experts to refine business KPIs and collaborate with Data and MLOps engineers within an Agile/SCRUM framework. Key projects include rule-based and NER-based data anonymization, creating privacy-preserving business ontologies and contributing to developing a privacy-preserving framework for synthetic data evaluation from external vendors. I regularly provide updates, create dashboards to assist business and legal stakeholders and present my projects to non-technical audiences.

Data Governance: In supporting KPN's move toward a decentralized data governance model, specifically a data mesh, I lead the formulation and implementation of data contracts that integrate privacy considerations from the beginning, aiding in GDPR compliance monitoring. This includes drafting functional and non-functional requirements, policies and data delivery terms aligned with KPN's internal data products to serve all stakeholders effectively.

15/10/2020 PhD Candidate

Jheronimus Academy of Data Science (JADS)

Main Role: I am affiliated with the Jheronimus Academy of Data Engineering (JADE) Lab. As a PhD candidate, I conduct scientific research on privacy engineering and privacy-preserving AI at the Dutch Telco provider KPN. Additionally, I am involved in various Horizon Europe-funded projects, including Infrastructure as Code (SODALITE H2020, RADON H2020), EU maritime trafficking intelligence (MARIT-D), and the development of integrated healthcare models for dementia and frailty prevention (COMFORTAge). My work results in publications in top-tier (Q1) journals and presentations at rank A conferences in privacy engineering and privacy-preserving AI.

Education: I teach three Master's level courses at JADS in the Data Science and Business Entrepreneurship program: Data Forensics, Natural Language Processing (NLP) and Data Intrapreneurship in Action. I have co-supervised three Master's students on theses related to privacy-preserving AI and text analytics.

Community Engagement: Engaging with the JADS community is key to my role. I actively participate in the JADS PhD Board, having joined in October 2022. I organize various community-building activities for PhD candidates and deliver public talks about my research at JADS events.

15/10/2019 - Data Scientist / Research Assistant

31/03/2020 Tilburg University

Source control tools:

GitHub, BitBucket, Mercurial

I conducted academic research in Infrastructure as Code (IaC), focusing on practical aspects including data collection, verification, cleaning, preparation, and pre-processing. This progressed to the development, evaluation, tuning, and deployment of a deep learning model. The culmination of my efforts is an academic paper accepted by the Workshop on Machine Learning Techniques for Software Quality Evaluation (MaLTaSQuE), showcasing research that applies Deep Learning (CNN) to identify name-based bugs in IaC.

This project is funded by the European Commission under grant nos. 825480 (SODALITE H2020) and 825040 (RADON H2020).

01/10/2016 - Data Engineer
17/08/2018 Mouzenidis Group

01/10/2014 - Full Stack Developer
30/09/2016 Mouzenidis Group

10/04/2013 - Junior Web Developer
30/09/2014 Mouzenidis Travel

Academic publications

Google Scholar Profile: <https://scholar.google.com/citations?user=VZOsMdAAAAAJ>

Education

15/10/2020 PhD in Data Engineering and AI for Privacy
Jheronimus Academy of Data Science

27/08/2018 - MSc Data Science & Entrepreneurship
10/07/2020 Jheronimus Academy of Data Science

01/10/2006 - BSc in Applied Informatics
18/07/2012 University of Macedonia

Languages

Mother tongues:

Greek, Serbian.

Other languages:

English: IELTS, Proficiency in English (University of Michigan).
Dutch, Russian: Basic communication skills / working knowledge.

Organizational Competencies

Creating and keeping deadlines	Project management	Teamwork
Goal setting and meeting goals	Creative thinking	Strategic thinking

Other interests

Artificial intelligence ethics enthusiast.
Jogging, playing football and basketball, cooking and travelling.